

Risk Control Matrix (RCM)

Model Name: HC Manuscript Evaluator

Model ID: HC-QG-001

Tier: Tier 1 (High Risk)

Version: 4.0

Date: April 2026

Purpose: This document details the key risks associated with the HC Manuscript Evaluator model, the controls designed to mitigate them, and the specific, actionable procedures for testing and responding to control failures. It serves as the primary operational playbook for risk management and the central evidence artifact for auditors.

Risk ID	Risk & Inherent Risk	Control ID	Control Description & Type	Control Owner	Testing Procedure	Failure Threshold & Response	Residual Risk
R-01	Bias/Fairness: Model produces systematically biased scores based on author-identifying information. Inherent: High	C-01.1	Monthly Bias Monitoring [Detective]	MRM Lead	Monthly, MRM runs 20 standard test queries with demographics redacted. Results are compared to an unredacted baseline and disparate impact ratio (DIR) is calculated.	Threshold: Avg score difference > 0.5 points OR DIR outside 0.8–1.25. Response: Formal MRM review initiated within 5 business days.	Medium
		C-01.2	Standardized Prompt Templates [Preventive]	Manager, Editorial Systems	Annual review of prompt neutrality and effectiveness by MRM. System validation confirms 100% of API calls use approved prompt template hashes.	Threshold: Any use of a non-approved prompt in production. Response: Incident escalated to Editorial Systems; user access may be temporarily suspended pending review and mandatory retraining.	
R-02	Accuracy/Performance: Model fabricates ("hallucinates") non-existent book titles. Inherent: High	C-02.1	Mandatory Checklist Verification of Comparable Titles [Preventive]	Acquisitions Editor	Editors are required to verify all AI-generated comparable titles using external sources and document completion via checklist prior to submission.	Threshold: Control effectiveness validated through C-02.2.	Low
		C-02.2	Checklist Adherence Audit [Detective]	MRM Lead	Monthly, MRM audits a 10% random sample of completed checklists to confirm all verification steps were completed and documented	Threshold: Any audited checklist found incomplete. Response: Immediate remedial training for the editor and retrospective review of their last 20 submissions	

		C-02.3	Hallucination Rate Monitoring [Detective]	MRM Lead	Monthly, a random sample of 10% of all submissions (minimum 50) is audited. Fabricated comparable titles are identified and counted.	Threshold: Hallucination rate > 5%. Response: Immediate model suspension, root cause analysis, and prompt/model remediation prior to reactivation.	
R-03	Legal/Compliance: Model reproduces copyrighted text, creating legal exposure. Inherent: High	C-03.1	Prompt Hardening & Checklist Flag [Preventive/Detective]	Mngr, Editorial Systems (Prompt); Acq. Editor (Checklist)	Prompt includes explicit instruction to avoid verbatim reproduction. Editors must flag any output resembling copyrighted or unoriginal text in the checklist.	Threshold: Any confirmed instance of verbatim, unattributed reproduction. Response: Triggers P1 incident and immediate escalation to Legal Compliance for review and containment.	Medium
R-04	Operational/Human Factor: Editors develop automation bias and over-rely on the AI. Inherent: High	C-04.1	Override & Checklist Monitoring [Detective/Corrective]	Editorial Director	Monthly, system tracks (1) Human Override Rate and (2) Checklist Completion Rate across all users..	Threshold: Checklist completion < 100% OR override rate outside 30%–70% range for two consecutive months. Response: Mandatory team calibration session and targeted retraining with MRM	Low
R-05	Governance/Compliance: Lack of audit trail prevents reconstruction of decisions. Inherent: High	C-05.1	Immutable API Audit Logging [Preventive]	Manager, Editorial Systems	All API transactions log timestamp, editor ID, prompt hash, and output hash to a write-only datastore. Quarterly audits validate completeness.	Threshold: > 0.1% of transactions missing in audit. Response: High-priority remediation ticket with 48-hour SLA and incident review.	Low

Conclusion:

This Risk Control Matrix demonstrates a layered control framework combining preventive, detective, and corrective mechanisms to reduce high inherent risks to acceptable residual levels. Each control includes defined ownership, measurable testing procedures, and explicit failure thresholds with corresponding response actions. As a result, this document functions as both an operational governance tool and a complete audit-ready artifact supporting ongoing model risk management.